Asma Gannar

Final-Year Software Engineering Student | AI & Cybersecurity
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Profile

Final-year Software Engineering student interested in **Cybersecurity, Artificial Intelligence, and Machine Learning**. Experience developing AI-powered applications and backend services, with a background in software quality, data processing, and debugging. Interested in applying software engineering and AI techniques to build **secure, reliable, and intelligent systems**.

Technical Skills

Cybersecurity: Network Security, Web Application Security, Identity & Access Management, Security Monitoring & Incident Response, Cryptography, Burp Suite
AI & Machine Learning: Machine Learning, LLMs, Retrieval-Augmented Generation (RAG), Prompt Engineering, Data Processing
Programming Languages: Python, TypeScript, JavaScript, Java, C, SQL
Frameworks & Technologies: FastAPI, Streamlit, React, Next.js, Node.js, Spring Boot, Ollama, AWS Bedrock
Databases & Tools: PostgreSQL, MySQL, MongoDB, ChromaDB, Supabase, Git, GitLab, GitHub Actions, Docker, Linux, CI/CD
Software Engineering & Quality: Software Architecture, REST APIs, Software Testing, Validation, Debugging, Reliability Analysis

Professional Experience

Sopra HR Software Tunis, Tunisia
Software Engineering Intern *Jul. 2026 – Aug. 2026*
- Worked on a **TypeScript-based Git branch analysis system** within a software migration platform, focusing on data processing and automated reporting.
- Audited core processing pipelines and identified **10 data-quality issues** involving file renames, deletions, binary files, and conflict analysis.
- Implemented, tested, and reviewed fixes, improving the accuracy and reliability of analysis results.
- Evaluated an **AWS Bedrock AI integration** to generate software engineering insights from structured repository data.

2M Stoneware Nabeul, Tunisia
Web Development Intern *Jul. 2024*
- Developed responsive features for a React-based e-commerce application, contributing to application workflows, data handling, and debugging.

B2M Company Hammamet, Tunisia
Web Development Intern *Jul. – Aug. 2023*
- Contributed to web application development through frontend implementation, debugging, and software problem-solving.

Selected Projects

CyberLibrarian – RAG Cybersecurity Assistant (Team Project) *Python, FastAPI, ChromaDB, Docker, Render*
- Developed a RAG-based assistant that answers cybersecurity questions with citations from OWASP, MITRE ATT&CK, and NIST resources.
- Implemented hybrid retrieval combining BM25 keyword search and vector search, with reciprocal rank fusion, query rewriting, and cross-encoder reranking.
- Integrated GPT-4o-mini to generate context-aware answers, with source validation and fallback handling for unsupported queries.
- Developed the FastAPI backend and web interface, added automated testing, privacy-conscious logging, and rate limiting, and deployed the application using Docker on Render.

XAI Internship Management Platform – Orange ISS *React, Python, REST APIs*
- Developed a platform integrating Python services, REST APIs, and AI-based CV analysis to support internship application management.
- Contributed to automated data processing workflows, React interfaces, and backend integration.

Gruup – Multi-Tenant SaaS Platform (gruup.dev) *Next.js, React, TypeScript, Supabase*
- Co-developed a full-stack multi-tenant platform featuring **role-based access control** and structured data management.
- Implemented member management, attendance, scheduling, billing, analytics, and reporting features, contributing to application architecture and data validation.

Education

Mediterranean Institute of Technology (MedTech), SMU Tunis, Tunisia
Engineering Degree in Software Engineering *2022 – 2027*

Certifications

Full-Stack Development – Udemy | Agile Project Management – Google